Back matter

Acronyms and abbreviations

Abbreviation	Expansion
ACC	anterior cingulate cortex
ACTH	adrenocorticotropic hormone
AI	artificial intelligence
AMPA	α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid
ANS	autonomic nervous system
ATP	adenosine triphosphate
BBB	blood–brain barrier
BOLD	blood-oxygen-level-dependent
CBT	cognitive behavioural therapy
CBT-I	cognitive behavioural therapy for insomnia
CB1	cannabinoid receptor type 1
CRH	corticotropin-releasing hormone
CSF	cerebrospinal fluid
CT	computed tomography
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition
EEG	electroencephalography
FDA	U.S. Food and Drug Administration
fMRI	functional magnetic resonance imaging
GABA	gamma-aminobutyric acid
GR	glucocorticoid receptor
GWAS	genome-wide association study
HPA	hypothalamic–pituitary–adrenal
HRV	heart rate variability
ICD-11	International Classification of Diseases, 11th Revision
IL-6	interleukin-6
LTD	long-term depression
LTP	long-term potentiation

Abbreviation	Expansion
MRI	magnetic resonance imaging
NCTSN	National Child Traumatic Stress Network
NIMH	National Institute of Mental Health
NMDA	N-methyl-D-aspartate
NPY	neuropeptide Y
PET	positron emission tomography
PFC	prefrontal cortex
PTSD	post-traumatic stress disorder
PVN	paraventricular nucleus
REM	rapid eye movement
rTMS	repetitive transcranial magnetic stimulation
SCN	suprachiasmatic nucleus
SNP	single nucleotide polymorphism
SSRI	selective serotonin reuptake inhibitor
TNF	tumour necrosis factor
VTA	ventral tegmental area
WHO	World Health Organization

Appendix A. Critical appraisal template

Use this template for a single study. Record the source, then answer each prompt in a short paragraph. A completed template is a record of reasoning, not a score.

Section	Prompts to answer
Citation and access	Full reference; did you read the full text, an abstract, or a summary?
Question	What question does the study ask? Is it a question about prevalence, mechanism, efficacy, experience or implementation?
Design	What design was used? Is it capable of answering the question?
Sample	Who was studied, how were they selected, and who was excluded?
Measures	What constructs were measured, with which instruments, and what are their reliability and validity?
Comparison	What is the comparison group or condition, and is it appropriate?
Analysis	Are the analyses prespecified and appropriate? Is multiple testing handled?
Effect and precision	What is the effect size, with what confidence interval? Is it clinically meaningful?
Bias	What selection, measurement, attrition and reporting biases are plausible?
Confounding	Which alternative explanations remain?
Applicability	To whom, in what setting, does this result plausibly apply?
Conclusion	What does the study support, what does it not support, and what would reduce uncertainty?

Appendix B. Trauma-informed assessment checklist

This checklist supports teaching and supervision. It is not a validated instrument and does not replace local protocols.

Domain	Checks
Safety and setting	Private space; interpreter if needed; no immediate danger; exit route; planned ending time
Consent and confidentiality	Purpose of questions; limits of confidentiality; storage of records; right to decline
History	What happened; when; for how long; who was involved; childhood and adult adversity; current stressors
Symptoms	Intrusion; avoidance; negative cognition and mood; arousal; dissociation; sleep; substance use
Risk	Suicide; self-harm; harm to others; safeguarding; access to means; protective factors
Health	Relevant medical conditions; medication; pain; sleep disorder; pregnancy; neurological red flags
Culture and language	Idioms of distress; explanatory models; language of assessment; role of family
Strengths	Supports; coping; values; previous helpful treatment
Formulation	Integrated explanation shared with the person and open to revision
Plan	Immediate safety; treatment; referral; review date; what to do in a crisis

Appendix C. Research protocol outline

A protocol for an educational project should be reviewed by the relevant ethics committee before data collection. This outline indicates the sections such a document normally contains; it is not itself an approved protocol.

Section	Content
Title and version	Short title; version number; date; responsible investigator
Background	What is known; what is uncertain; why this question matters
Question and objectives	Primary question; secondary questions; hypotheses, if any
Design	Design and its justification; comparators; duration
Participants	Inclusion and exclusion criteria; recruitment; consent procedure; withdrawal
Measures	Constructs; instruments; reliability and validity; translation
Procedure	What participants do; time burden; setting; adaptations
Analysis	Primary analysis; missing data; sensitivity analyses; power or information power
Ethics	Risks; benefits; safeguards; confidentiality; data protection; incidental findings
Data management	Storage; access; retention; sharing; destruction
Dissemination	Community feedback; publication; authorship; data availability
Timeline and resources	Milestones; staff; costs; contingencies

Appendix D. Statistical and methodological terms

Bias. A systematic error that distorts an estimate. Bias differs from random error, which is reduced by larger samples.

Confidence interval. A range of values consistent with the observed data at a stated level of confidence. It conveys precision and is more informative than a significance test alone.

Confounding. A third variable associated with both exposure and outcome that distorts their apparent relationship.

Effect size. The magnitude of a difference or association, expressed in standardised or natural units.

Heterogeneity. Variation in effect estimates across studies or subgroups beyond what sampling error would predict.

Intention-to-treat. Analysis of participants according to the group to which they were allocated, regardless of the treatment received.

Power. The probability of detecting an effect of a given size if it exists. Underpowered studies produce unstable estimates.

Preregistration. Public registration of hypotheses, outcomes and analyses before data collection or analysis.

Sensitivity analysis. Reanalysis under different assumptions to test whether conclusions are robust.

Research resources

The following organisations publish material that is useful for teaching and for orientation to services. Resources change; verify the current page, date and jurisdiction before relying on any item. Listing does not imply endorsement.

Resource	Purpose
National Institute of Mental Health (nimh.nih.gov)	Public and professional information on PTSD and related conditions
National Child Traumatic Stress Network (nctsn.org)	Resources on child trauma, complex trauma and service models
World Health Organization (who.int)	International classifications, guidance and global mental health material
PubMed (pubmed.ncbi.nlm.nih.gov)	Bibliographic database for biomedical and psychological literature
Cochrane Library (cochranelibrary.com)	Systematic reviews of health interventions
National Institute for Health and Care Excellence (nice.org.uk)	Clinical guidance and evidence reviews for England and Wales
American Psychological Association (apa.org)	Ethics guidance, publication standards and professional resources

Laboratory and classroom teaching protocols

These protocols describe teaching activities only. They do not authorise clinical investigation, and they do not produce diagnostic information. Each requires institutional approval, supervision by a qualified teacher, informed consent from participants, and a risk assessment. Where a protocol involves biological samples, local biosafety, storage and disposal rules apply, and the assay manufacturer's instructions govern the procedure.

Protocol 1. Salivary cortisol sampling (educational)

Aim. To demonstrate diurnal variation and the practical constraints of endocrine measurement.

Materials. Salivary collection devices approved for human use; labels; timed reminders; cold storage; assay arranged through an approved laboratory.

Procedure. Students review the sampling schedule specified by the assay and the study question; record sleep, food, caffeine, nicotine, medication and exercise; collect samples at the approved times; store and transport them according to the laboratory's instructions.

Analysis. Plot individual values against time. Discuss why a single sample cannot characterise a person's stress physiology.

Limits. Values depend on time of day, posture, recent activity, oral health and assay method. Results are not interpreted clinically.

Protocol 2. Heart rate variability demonstration

Aim. To show how autonomic regulation can be indexed indirectly.

Materials. A validated recording device; electrodes or photoplethysmography sensor; analysis software with documented artefact handling.

Procedure. Record resting and paced-breathing conditions; inspect the trace for artefacts; compute time-domain and frequency-domain measures with the software's stated settings.

Analysis. Compare conditions within participants and discuss the influence of posture, breathing rate, fitness and medication.

Limits. Consumer devices vary in validity; short recordings are unstable; group averages do not describe individuals.

Protocol 3. Conditioning and extinction demonstration

Aim. To illustrate acquisition, extinction and renewal using mild stimuli only.

Materials. Computer task with neutral images and brief, non-painful sounds; structured instructions; debriefing.

Procedure. Participants learn that one image predicts a sound; the sound is then omitted; the image is presented in a new context. Subjective expectancy ratings and simple reaction times are recorded.

Analysis. Plot expectancy across phases and discuss renewal.

Limits. Mild laboratory conditioning differs from traumatic learning. No aversive stimulation is used, and participation is voluntary and withdrawable.

Protocol 4. Neuroanatomy practical

Aim. To identify the structures discussed in Chapters 2–4.

Materials. Anatomical models, plastinated sections or approved digital atlases.

Procedure. Students locate the hippocampus, amygdala, insula, anterior cingulate, hypothalamus, brainstem nuclei and ventricular system; they trace the course of the HPA axis and the main noradrenergic projection.

Analysis. Short structured identification exercise with a marking guide.

Limits. Models simplify anatomy; imaging appearances differ from specimens.

Professional standards and scope of practice

This textbook supports doctoral study of trauma neurobiology. It does not confer clinical competence, prescribing authority, imaging privileges, laboratory certification or legal authority to assess trauma in forensic settings.

Students and practitioners should work within their competence and their jurisdiction's law; seek supervision when a case exceeds that competence; obtain informed consent and explain the limits of confidentiality; protect identifiable data; recognise and declare conflicts of interest; and refer for medical assessment when symptoms, history or examination suggest a condition requiring it.

Research with trauma-affected people requires ethical review, valid consent, attention to power and dependency, referral pathways for distress, and arrangements for sharing findings with participants and communities. Authorship should reflect substantive contribution. Findings should be reported honestly, including those that are negative or inconvenient.

Where this book and local law, professional regulation or institutional policy differ, the latter takes precedence.

Further reading

The following works are cited in the chapters and are useful starting points. Annotations indicate scope, not endorsement.

- **Shonkoff and Garner (2012)** — the American Academy of Pediatrics technical report on childhood adversity and toxic stress; a concise framing of developmental and policy implications.
- **Teicher and Samson (2016)** — a review of the neurobiological correlates of childhood maltreatment, with attention to sensitive periods and methodological limits.
- **McEwen (1998)** — the classic account of allostasis and allostatic load, still the clearest short statement of the framework.
- **Ulrich-Lai and Herman (2009)** — neural regulation of endocrine and autonomic stress responses.
- **Herman et al. (2016)** — detailed physiology of the HPA axis, including feedback and circadian control.
- **LeDoux (2000)** and **Maren (2001)** — complementary accounts of fear circuitry and Pavlovian conditioning.
- **Milad and Quirk (2012)** — extinction as a translational model, with a clear account of relapse phenomena.
- **Nader et al. (2000)** — the reconsolidation experiment that reframed memory stability.
- **Vervliet et al. (2013)** — a critical review of extinction and relapse; useful for judging clinical claims.
- **Craske et al. (2014)** — inhibitory learning and the practical optimisation of exposure.
- **Ehlers and Clark (2000)** — the cognitive model that organises much trauma-focused therapy.
- **Lanius et al. (2010)** — the dissociative subtype and its neurobiological basis.
- **Etkin and Wager (2007)** — meta-analytic imaging findings across anxiety disorders.
- **Germain (2013)** — sleep disturbance as a central feature of PTSD.
- **Hori and Kim (2019)** — inflammation in PTSD, with appropriate caution about causality.
- **Yehuda et al. (2015)** — an authoritative overview of PTSD, from epidemiology to treatment.
- **Felitti et al. (1998)** — the original adverse childhood experiences study.

- **Kessler et al. (1995)** — the National Comorbidity Survey estimates of trauma exposure and PTSD.
- **Page et al. (2021)** — the PRISMA 2020 reporting standard for systematic reviews.

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